

# Math 237B Final

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March 15, 2026

## 1

**Problem 1.** Let  $X \xrightarrow{f} Y \xrightarrow{g} Z$  be morphisms of schemes such that  $g \circ f$  is proper,  $g$  is quasi-projective and  $f$  is surjective. Prove or disprove that  $g$  is proper.

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*Solution.* We'll prove that  $g$  is proper. We'll prove relevant statements in the following order:

- (a)  $g$  is separated, and  $f$  is proper.
- (b)  $g$  is of finite type.
- (c)  $g$  is universally closed.

Finally, use these results we can conclude that  $g$  is proper.

### Proof of (a):

Since  $g : Y \rightarrow Z$  is quasi-projective, there exists some scheme  $Y'$ , together with open immersion  $\iota : Y \hookrightarrow Y'$  and projective morphism  $\pi : Y' \rightarrow Z$ , such that the following commutative diagram holds:

$$\begin{array}{ccc} Y & \xhookrightarrow{\iota} & Y' \\ g \downarrow & \swarrow \pi & \\ Z & & \end{array}$$

Which, recall that any open immersion is separated, and projective morphisms are proper (in particular separated), so  $g = \pi \circ \iota$  is also separated.

As a consequence, since  $g \circ f$  is proper, while  $g$  is separated, we concluded that  $f$  is proper also.

### Proof of (b):

Let's show  $g$  is of finite type: Using the diagram in part (a), since  $\pi : Y' \rightarrow Z$  is projective, it is proper (hence of finite type), so for any open affine subset  $U \subseteq Z$  (say  $U \cong \text{Spec}(A)$ ), such that  $\pi^{-1}(U) = \bigcup_{i=1}^n V_i$ , where each  $V_i$  is affine open subset, and  $V_i \cong \text{Spec}(B_i)$  for some  $B_i$  that's finitely generated  $A$ -algebra.

Then, with open immersion  $\iota : Y \hookrightarrow Y'$ , one has  $\iota^{-1}(\pi^{-1}(U)) = Y \cap \pi^{-1}(U) = \bigcup_{i=1}^n Y \cap V_i$ . Since  $Y$  is open in  $Y'$ , each  $Y \cap V_i$  is an open subset of  $V_i$  under its subspace topology, hence each  $Y \cap V_i = \bigcup_{j \in J_i} D(f_{ij})$  for some  $f_{ij} \in B_i$  (running over some index set  $J_i$ ). Because each  $D(f_{ij}) \cong \text{Spec}\left((B_i)_{f_{ij}}\right)$  (which  $(B_i)_{f_{ij}}$  is a finitely generated  $B_i$ -algebra), then with  $B_i$  being a finitely generated  $A$ -algebra,  $(B_i)_{f_{ij}}$  is a finitely generated  $A$ -algebra also.

As a result, one has  $g^{-1}(U) = \iota^{-1}(\pi^{-1}(U)) = \bigcup_{i=1}^n Y \cap V_i = \bigcup_{i=1}^n \bigcup_{j \in J_i} D(f_{ij})$ , where each  $D(f_{ij}) \cong \text{Spec}\left((B_i)_{f_{ij}}\right)$  has  $(B_i)_{f_{ij}}$  being a finitely generated  $A$ -algebra.

Finally, since  $g \circ f$  is finite type, it's quasi-compact, hence with affine open subset  $U \subseteq Z$ , one has  $(g \circ f)^{-1}(U) = f^{-1}(g^{-1}(U))$  being quasi-compact. However, since  $f$  is surjective, one has  $g^{-1}(U) =$

$f(f^{-1}(g^{-1}(U)))$ . Because it's an image of a quasi-compact set under continuous map,  $g^{-1}(U)$  is quasi-compact. Hence, the previous cover for  $g^{-1}(U)$ , say  $g^{-1}(U) = \bigcup_{i=1}^n \bigcup_{j \in J_i} D(f_{ij})$ , can be chosen as finite. This shows that  $g$  is of finite type.

**Proof of (c):**

To prove  $g$  is universally closed, for any morphism of schemes  $h : Z' \rightarrow Z$  consider the following commutative diagram of multiple fibre products:

$$\begin{array}{ccc}
 (Z' \times_Z Y) \times_Y X & \xrightarrow{h''} & X \\
 \downarrow f' & & \downarrow f \\
 Z' \times_Z Y & \xrightarrow{h'} & Y \\
 \downarrow g' & & \downarrow g \\
 Z' & \xrightarrow{h} & Z
 \end{array}$$

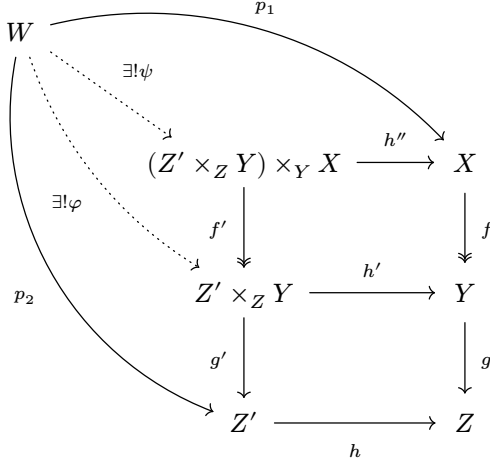
Which:

- $g' : Z' \times_Z Y \rightarrow Z'$  denotes the base change of  $g : Y \rightarrow Z$  by  $h : Z' \rightarrow Z$ .
- $f' : (Z' \times_Z Y) \times_Y X \rightarrow Z' \times_Z Y$  denotes the base change of  $f : X \rightarrow Y$  by  $h' : Z' \times_Z Y \rightarrow Y$  (the pullback of  $h : Z' \rightarrow Z$ ), which the surjectivity of  $f$  ensures the surjectivity of  $f'$ . And, since part (a) shows  $f$  is proper (which is universally closed), then  $f'$  is closed.

Now, we ought to show  $(Z' \times_Z Y) \times_Y X$  together with  $g' \circ f'$  and  $h''$  satisfies the universal property of the fibre product  $Z' \times_Z X$ . For any scheme  $W$  together with two morphisms  $p_1 : W \rightarrow X$  and  $p_2 : W \rightarrow Z'$ , such that  $(g \circ f) \circ p_1 = h \circ p_2$ . Then, the universal property of  $Z' \times_Z Y$  guarantees a unique morphism of schemes  $\varphi : W \rightarrow Z' \times_Z Y$ , such that  $p_2 = g' \circ \varphi$ , and  $f \circ p_1 = h' \circ \varphi$ , as follow:

$$\begin{array}{ccc}
 W & \xrightarrow{p_1} & X \\
 \downarrow \exists! \varphi & & \downarrow f' \\
 (Z' \times_Z Y) \times_Y X & \xrightarrow{h''} & X \\
 \downarrow f' & & \downarrow f \\
 Z' \times_Z Y & \xrightarrow{h'} & Y \\
 \downarrow g' & & \downarrow g \\
 Z' & \xrightarrow{h} & Z
 \end{array}$$

Furthermore, with the universal property of  $(Z' \times_Z Y) \times_Y X$ , the equality  $f \circ p_1 = h' \circ \varphi$  guarantees a unique morphism of schemes  $\psi : W \rightarrow (Z' \times_Z Y) \times_Y X$ , such that  $\varphi = f' \circ \psi$  and  $p_1 = h'' \circ \psi$ , as follow:



This proves both the existence and the uniqueness of the morphism  $\psi : W \rightarrow (Z' \times_Z Y) \times_Y X$ , such that  $p_2 = g' \circ \varphi = (g' \circ f') \circ \psi$ , and  $p_1 = h'' \circ \psi$  (as any  $\psi' : W \rightarrow (Z' \times_Z Y) \times_Y X$  with such property is enforced to be equal to  $\psi$ , based on individual fibre product's property). So,  $(Z' \times_Z Y) \times_Y X$  indeed satisfies the universal property of  $Z' \times_Z X$ , showing that  $g' \circ f'$  is the base change of the morphism  $(g \circ f) : X \rightarrow Z$ . Which, since  $g \circ f$  is proper,  $g' \circ f'$  is closed.

Finally, with  $f'$  and  $g' \circ f'$  being closed, and  $f'$  being surjective, we claim that  $g'$  is closed:

For any closed subset  $V \subseteq Z' \times_Z Y$ , by the continuity of  $f'$ , the subset  $(f')^{-1}(V) \subseteq (Z' \times_Z Y) \times_Y X$  is closed. Hence,  $(g' \circ f')((f')^{-1}(V)) \subseteq Z'$  is closed (by closeness of  $g' \circ f'$ ).

Also, since  $f'$  is surjective, one simply has  $f'((f')^{-1}(V)) = V$ . Hence, the above set equality shows that  $(g' \circ f')((f')^{-1}(V)) = g'(V) \subseteq Z'$  is closed. This proves that  $g'$  is closed.

With  $g'$  being based change of  $g$  over arbitrary morphism of schemes  $h : Z' \rightarrow Z$ , this shows that  $g$  is universally closed.

With all information above, we can conclude that  $g$  is separated, finite type, and universally closed, hence proper.

## 2

**Problem 2.** Let  $f : X \rightarrow S$  and  $g : Y \rightarrow S$  be projective morphisms. Prove or disprove that the projection map  $X \times_S Y \rightarrow S$  is projective.

*Solution.* We'll show that  $X \times_S Y \rightarrow S$  is projective. We'll prove the following two statements:

- (a) Base change of projective morphisms are projective.
- (b) Composition of projective morphisms are projective.

Then, the two will help conclude the desired result.

### Proof of (a):

First, we'll show that base change preserves projective spaces: More precisely, given any  $n \in \mathbb{N}$ , with morphisms  $g : Y \rightarrow S$ , the projective  $n$ -space over  $S$ ,  $\pi : \mathbb{P}_S^n \rightarrow S$  has fibre product  $Y \times_S \mathbb{P}_S^n$  satisfies the universal property of  $\mathbb{P}_Y^n$ .

Indeed, with the definition of  $\mathbb{P}_S^n := S \times_{\text{Spec}(\mathbb{Z})} \mathbb{P}_{\mathbb{Z}}^n$ , we form the following commutative diagram:

$$\begin{array}{ccccc}
 Y \times_S \mathbb{P}_S^n & \xrightarrow{g'} & \mathbb{P}_S^n & \xrightarrow{s'} & \mathbb{P}_{\mathbb{Z}}^n \\
 \pi' \downarrow & & \downarrow \pi & & \downarrow \varphi \\
 Y & \xrightarrow{g} & S & \xrightarrow{s} & \text{Spec}(\mathbb{Z})
 \end{array}$$

Where, both small squares are fibre product diagrams. Also, the morphism  $s \circ g : Y \rightarrow \text{Spec}(\mathbb{Z})$  is the unique morphism based on the terminal property of  $\text{Spec}(\mathbb{Z})$  in the category of schemes.

Now, given any scheme  $W$  together with two morphisms  $p_1 : W \rightarrow \mathbb{P}_{\mathbb{Z}}^n$  and  $p_2 : W \rightarrow Y$ , such that  $\varphi \circ p_1 = (s \circ g) \circ p_2$ , then the universal property of  $\mathbb{P}_S^n$  generates a unique morphism  $\psi : W \rightarrow \mathbb{P}_S^n$ , such that  $p_1 = s' \circ \psi$ , and  $g \circ p_2 = \pi \circ \psi$ , as follow:

$$\begin{array}{c}
 \begin{array}{ccccc}
 W & & & & \\
 \curvearrowright^{p_1} & & & & \\
 & \searrow^{\exists! \psi} & & & \\
 & & Y \times_S \mathbb{P}_S^n & \xrightarrow{g'} & \mathbb{P}_S^n & \xrightarrow{s'} & \mathbb{P}_{\mathbb{Z}}^n \\
 & & \pi' \downarrow & & \downarrow \pi & & \downarrow \varphi \\
 & & Y & \xrightarrow{g} & S & \xrightarrow{s} & \text{Spec}(\mathbb{Z}) \\
 & \searrow^{p_2} & & & & & \\
 & & & & & & 
 \end{array}
 \end{array}$$

Then, with the universal property of  $Y \times_S \mathbb{P}_S^n$ , the second equality  $g \circ p_2 = \pi \circ \psi$  guarantees a unique morphism  $\rho : W \rightarrow Y \times_S \mathbb{P}_S^n$ , such that  $\psi = g' \circ \rho$ , and  $p_2 = \pi' \circ \rho$ , as follow:

$$\begin{array}{c}
 \begin{array}{ccccc}
 W & & & & \\
 \curvearrowright^{p_1} & & & & \\
 & \searrow^{\exists! \psi} & & & \\
 & \searrow^{\exists! \rho} & & & \\
 & & Y \times_S \mathbb{P}_S^n & \xrightarrow{g'} & \mathbb{P}_S^n & \xrightarrow{s'} & \mathbb{P}_{\mathbb{Z}}^n \\
 & & \pi' \downarrow & & \downarrow \pi & & \downarrow \varphi \\
 & & Y & \xrightarrow{g} & S & \xrightarrow{s} & \text{Spec}(\mathbb{Z}) \\
 & \searrow^{p_2} & & & & & \\
 & & & & & & 
 \end{array}
 \end{array}$$

Again, this proves both the existence and the uniqueness of the morphism  $\rho : W \rightarrow Y \times_S \mathbb{P}_S^n$ , such that  $p_1 = s' \circ \psi = (s' \circ g') \circ \rho$ , and  $p_2 = \pi' \circ \rho$ . Hence,  $Y \times_S \mathbb{P}_S^n$  satisfies the universal property of  $\mathbb{P}_Y^n := Y \times_{\text{Spec}(\mathbb{Z})} \mathbb{P}_{\mathbb{Z}}^n$ , showing  $Y \times_S \mathbb{P}_S^n \cong \mathbb{P}_Y^n$ .

Now, we'll prove that base change preserves projective morphisms, again by the universality argument: Given a projective morphism  $f : X \rightarrow S$ , and another morphism  $g : Y \rightarrow S$ , we know there exists a suitable  $n \in \mathbb{N}$ , such that there exists a closed immersion  $\iota : X \hookrightarrow \mathbb{P}_S^n$  and the projection  $\pi : \mathbb{P}_S^n \rightarrow S$ , such that  $f = \pi \circ \iota$ .

Which, based on the previous argument, we'll simply denote  $Y \times_S \mathbb{P}_S^n$  as  $\mathbb{P}_Y^n$ . Together with the previous diagrams, we have the following:

$$\begin{array}{ccccc}
 X \times_{\mathbb{P}_S^n} \mathbb{P}_Y^n & \xrightarrow{g''} & X & & \\
 \downarrow \iota' & & \downarrow \iota & & \\
 \mathbb{P}_Y^n & \xrightarrow{g'} & \mathbb{P}_S^n & \xrightarrow{s'} & \mathbb{P}_{\mathbb{Z}}^n \\
 \downarrow \pi' & & \downarrow \pi & & \downarrow \varphi \\
 Y & \xrightarrow{g} & S & \xrightarrow{s} & \text{Spec}(\mathbb{Z})
 \end{array}$$

Which, we took the fibre product  $X \times_{\mathbb{P}_S^n} \mathbb{P}_Y^n$ , and the morphism  $\iota' : X \times_{\mathbb{P}_S^n} \mathbb{P}_Y^n \hookrightarrow \mathbb{P}_Y^n$  is the pullback of the closed immersion  $\iota : X \hookrightarrow \mathbb{P}_S^n$ , hence  $\iota'$  is also a closed immersion.

Now, we claim that it satisfies the universal property of  $X \times_S Y$ : In general, the above proof of preserving projective spaces shows that successive fibre product preserves fibre product in the full generality. By modifying the labels / objects a bit, we indeed get a proof that  $X \times_{\mathbb{P}_S^n} \mathbb{P}_Y^n$  satisfies the universal property of  $X \times_S Y$  as fibre product, hence  $X \times_{\mathbb{P}_S^n} \mathbb{P}_Y^n \cong X \times_S Y$ .

Finally, by denoting  $X \times_{\mathbb{P}_S^n} \mathbb{P}_Y^n$  as  $X \times_S Y$ , the above diagram reduces to the following fibre product diagram:

$$\begin{array}{ccc}
 X \times_S Y & \xrightarrow{g''} & X \\
 \downarrow \iota' & & \downarrow \iota \\
 \mathbb{P}_Y^n & & \mathbb{P}_S^n \\
 \downarrow \pi' & & \downarrow \pi \\
 Y & \xrightarrow{g} & S
 \end{array}
 \quad \begin{array}{c} \curvearrowright f \end{array}$$

Which, the pullback of  $f$  is  $\pi' \circ \iota' : X \times_S Y \rightarrow Y$ , which is precisely a closed immersion into  $\mathbb{P}_Y^n$ , then a projection onto  $Y$ . This shows that the pullback of  $f$  is still projective, hence projective morphisms are preserved under base change.

### Proof of (b):

Let  $\rho : T \rightarrow U$  and  $\varphi : U \rightarrow V$  be two projective morphisms, then there exists  $n, m \in \mathbb{N}$ , such that there exists open immersion  $\iota_T : T \hookrightarrow \mathbb{P}_U^n$  and  $\iota_U : U \hookrightarrow \mathbb{P}_V^m$ , together with projections  $\pi_{U,n} : \mathbb{P}_U^n \rightarrow U$  and  $\pi_{V,m} : \mathbb{P}_V^m \rightarrow V$ , such that the following diagram commutes:

$$\begin{array}{ccccc}
 \mathbb{P}_U^n & & \mathbb{P}_V^m & & \\
 \downarrow \iota_T & \searrow \pi_{U,n} & \downarrow \iota_U & \searrow \pi_{V,m} & \\
 T & \xrightarrow{\rho} & U & \xrightarrow{\varphi} & V
 \end{array}$$

Now, let's use some information from part (a): Recall that given any morphism  $Y \rightarrow S$  and the projection  $\mathbb{P}_S^n \rightarrow S$ , the fibre product  $Y \times_S \mathbb{P}_S^n \cong \mathbb{P}_Y^n$  (and it's true for arbitrary  $Y, S$  and  $n \in \mathbb{N}$ , as the proof is universal); then, from the above diagram, consider  $\varphi : U \rightarrow V$  as the composition  $U \xrightarrow{\iota_U} \mathbb{P}_V^m \xrightarrow{\pi_{V,m}} V$ , and consider the projection  $\pi_{V,n} : \mathbb{P}_V^n \rightarrow V$ . Notice by the previous statement, we can construct the projective space  $\mathbb{P}_{\mathbb{P}_V^m}^n \cong \mathbb{P}_V^m \times_V \mathbb{P}_V^n$ , as follow:

$$\begin{array}{ccccc} & & \mathbb{P}_{\mathbb{P}_V^m}^n & \xrightarrow{(\pi_{V,m})'} & \mathbb{P}_V^n \\ & & \downarrow (\pi_{V,n})' & & \downarrow \pi_{V,n} \\ U & \xhookrightarrow{\iota_U} & \mathbb{P}_V^m & \xrightarrow{\pi_{V,m}} & V \end{array}$$

Which, the same argument shows that  $\mathbb{P}_U^n \cong U \times_{\mathbb{P}_V^m} \mathbb{P}_{\mathbb{P}_V^m}^n$ , as follow:

$$\begin{array}{ccccc} \mathbb{P}_U^n & \xhookrightarrow{(\iota_U)'} & \mathbb{P}_{\mathbb{P}_V^m}^n & \xrightarrow{(\pi_{V,m})'} & \mathbb{P}_V^n \\ \downarrow \pi_{U,n} & & \downarrow (\pi_{V,n})' & & \downarrow \pi_{V,n} \\ U & \xhookrightarrow{\iota_U} & \mathbb{P}_V^m & \xrightarrow{\pi_{V,m}} & V \end{array}$$

Which, notice that  $(\iota_U)' : \mathbb{P}_U^n \hookrightarrow \mathbb{P}_{\mathbb{P}_V^m}^n$  is a base change of the closed immersion  $\iota_U : U \hookrightarrow \mathbb{P}_V^m$ , hence is still a closed immersion; composing with the closed immersion  $\iota_T : T \hookrightarrow \mathbb{P}_U^n$ , we get a closed immersion  $(\iota_U)' \circ \iota_T : T \hookrightarrow \mathbb{P}_{\mathbb{P}_V^m}^n \cong \mathbb{P}_V^m \times_V \mathbb{P}_V^n$ , and with the projections  $\pi_{V,n} \circ (\pi_{V,m})' : \mathbb{P}_{\mathbb{P}_V^m}^n \rightarrow V$ , they compose to be the original morphism  $T \xrightarrow{\varphi} U \rightarrow V$  (as  $(\pi_{V,n} \circ (\pi_{V,m})') \circ ((\iota_U)' \circ \iota_T) = (\pi_{V,n} \circ \iota_U) \circ (\pi_{U,n} \circ \iota_T) = \varphi \circ \rho$  based on the diagrams).

Hence, to prove the composition  $\varphi \circ \rho : T \rightarrow V$  is projective, it suffices to show that  $\mathbb{P}_{\mathbb{P}_V^m}^n \cong \mathbb{P}_V^m \times_V \mathbb{P}_V^n$  embeds into some  $\mathbb{P}_V^N$ , and the projection backs to  $\mathbb{V}$  recovers  $\varphi \circ \rho$ .

Let's first show that there exists a morphism  $p : \mathbb{P}_V^m \times_V \mathbb{P}_V^n \rightarrow \mathbb{P}_{\mathbb{Z}}^m \times_{\text{Spec}(\mathbb{Z})} \mathbb{P}_{\mathbb{Z}}^n$ . Consider the following diagram:

$$\begin{array}{ccccc} \mathbb{P}_V^m \times_V \mathbb{P}_V^n & \xrightarrow{(\pi_{V,m})'} & \mathbb{P}_V^n & \longrightarrow & \mathbb{P}_{\mathbb{Z}}^n \\ \downarrow (\pi_{V,n})' & & \downarrow \pi_{V,n} & & \downarrow \\ \mathbb{P}_V^m & \xrightarrow{\pi_{V,m}} & V & \searrow & \\ \downarrow & & & & \downarrow \\ \mathbb{P}_{\mathbb{Z}}^m & \longrightarrow & \text{Spec}(\mathbb{Z}) & & \end{array}$$

Which, with each „square“ (two are deformed a bit) being fibre squares, the above diagram commutes. Hence, by the universal property of  $\mathbb{P}_{\mathbb{Z}}^m \times_{\text{Spec}(\mathbb{Z})} \mathbb{P}_{\mathbb{Z}}^n$  guarantees a unique morphism  $P : \mathbb{P}_V^m \times_V \mathbb{P}_V^n \rightarrow \mathbb{P}_{\mathbb{Z}}^m \times_{\text{Spec}(\mathbb{Z})} \mathbb{P}_{\mathbb{Z}}^n$ .

Now, consider the Segre Embedding that is suggested in Hartshorne: Given  $\mathbb{P}_{\mathbb{Z}}^m = \text{Proj}(\mathbb{Z}[x_0, x_1, \dots, x_m])$  and  $\mathbb{P}_{\mathbb{Z}}^n = \text{Proj}(\mathbb{Z}[y_0, y_1, \dots, y_n])$ , we cover the fibre product  $\mathbb{P}_{\mathbb{Z}}^m \times_{\text{Spec}(\mathbb{Z})} \mathbb{P}_{\mathbb{Z}}^n$  by the open affine subsets  $D_+(x_i) \times_{\text{Spec}(\mathbb{Z})} D_+(y_j)$  for  $0 \leq i \leq m$  and  $0 \leq j \leq n$ . Which, with each  $D_+(x_i) \cong \text{Spec}\left(\mathbb{Z}\left[\frac{x_0}{x_i}, \frac{x_1}{x_i}, \dots, \frac{x_m}{x_i}\right]\right)$  and  $D_+(y_j) \cong \text{Spec}\left(\mathbb{Z}\left[\frac{y_0}{y_j}, \frac{y_1}{y_j}, \dots, \frac{y_n}{y_j}\right]\right)$ . Hence, their fibre product over  $\text{Spec}(\mathbb{Z})$  are given as follow:

$$\begin{aligned} D_+(x_i) \times_{\text{Spec}(\mathbb{Z})} D_+(y_j) &\cong \text{Spec}\left(\mathbb{Z}\left[\frac{x_0}{x_i}, \dots, \frac{x_m}{x_i}\right] \otimes_{\mathbb{Z}} \mathbb{Z}\left[\frac{y_0}{y_j}, \dots, \frac{y_n}{y_j}\right]\right) \\ &\cong \text{Spec}\left(\mathbb{Z}\left[\frac{x_0}{x_i}, \dots, \frac{x_m}{x_i}, \frac{y_0}{y_j}, \dots, \frac{y_n}{y_j}\right]\right) \end{aligned} \tag{2.1}$$

Then, with  $N := nm + n + m$ , consider  $\mathbb{P}_{\mathbb{Z}}^N = \text{Proj}(\mathbb{Z}[z_0, \dots, z_{ij}, \dots, z_N])$ . For any indeterminate  $z_{ij}$ , consider its fundamental open subset  $D_+(z_{ij}) = \text{Spec}(\mathbb{Z}[\frac{z_0}{z_{ij}}, \dots, \frac{z_N}{z_{ij}}])$ , and a ring homomorphism  $\psi_{ij} : \mathbb{Z}[\frac{z_0}{z_{ij}}, \dots, \frac{z_N}{z_{ij}}] \rightarrow \mathbb{Z}[\frac{x_0}{x_i}, \dots, \frac{x_m}{x_i}, \frac{y_0}{y_j}, \dots, \frac{y_n}{y_j}]$  as follow:

$$\psi_{ij}\left(\frac{z_{i'j'}}{z_{ij}}\right) = \frac{x_{i'} y_{j'}}{x_i y_j} \quad (2.2)$$

Notice that any  $\frac{x_{i'}}{x_i}$  satisfies  $\psi_{ij}\left(\frac{z_{i'j'}}{z_{ij}}\right) = \frac{x_{i'} y_{j'}}{x_i y_j} = \frac{x_{i'}}{x_i}$ , and any  $\frac{y_{j'}}{y_j}$  satisfies  $\psi_{ij}\left(\frac{z_{i'j'}}{z_{ij}}\right) = \frac{x_i y_{j'}}{x_i y_j} = \frac{y_{j'}}{y_j}$ . So, it maps onto all the generators of  $\mathbb{Z}[\frac{x_0}{x_i}, \dots, \frac{x_m}{x_i}, \frac{y_0}{y_j}, \dots, \frac{y_n}{y_j}]$ , showing  $\psi_{ij}$  is surjective.

Which, this defines a closed immersion  $\psi_{ij}^* : \text{Spec}(\mathbb{Z}[\frac{x_0}{x_i}, \dots, \frac{x_m}{x_i}, \frac{y_0}{y_j}, \dots, \frac{y_n}{y_j}]) = D_+(x_i) \times_{\text{Spec}(\mathbb{Z})} D_+(y_j) \hookrightarrow \text{Spec}(\mathbb{Z}[\frac{z_0}{z_{ij}}, \dots, \frac{z_N}{z_{ij}}]) = D_+(z_{ij})$  (as the ring homomorphism inducing it is surjective). Upon gluing the morphisms using the structure of  $\mathbb{P}_{\mathbb{Z}}^N$ , this guarantees a closed immersion  $\mathbb{P}_{\mathbb{Z}}^m \times_{\text{Spec}(\mathbb{Z})} \mathbb{P}_{\mathbb{Z}}^n \hookrightarrow \mathbb{P}_{\mathbb{Z}}^N$  (where  $N = nm + n + m$ ).

With the existence of Segre Embedding for projective spaces of  $\mathbb{Z}$ , say  $s : \mathbb{P}_{\mathbb{Z}}^m \times_{\text{Spec}(\mathbb{Z})} \mathbb{P}_{\mathbb{Z}}^n \hookrightarrow \mathbb{P}_{\mathbb{Z}}^{nm+n+m}$  (a closed immersion specifically), then with  $N := nm + n + m$  consider the following commutative diagram:

$$\begin{array}{ccc} \mathbb{P}_V^m \times_V \mathbb{P}_V^n & \xrightarrow{p} & \mathbb{P}_{\mathbb{Z}}^m \times_{\text{Spec}(\mathbb{Z})} \mathbb{P}_{\mathbb{Z}}^n \\ \downarrow \exists! s' & & \downarrow s \\ \mathbb{P}_V^N & \xrightarrow{\quad} & \mathbb{P}_{\mathbb{Z}}^N \\ \downarrow \pi_{V,N} & & \downarrow \\ V & \xrightarrow{\quad} & \text{Spec}(\mathbb{Z}) \end{array}$$

$\pi_{V,n} \circ (\pi_{V,m})'$

Then, the universality of  $\mathbb{P}_V^N = V \times_{\text{Spec}(\mathbb{Z})} \mathbb{P}_{\mathbb{Z}}^N$  guarantees a unique morphism  $s' : \mathbb{P}_V^m \times_V \mathbb{P}_V^n \rightarrow \mathbb{P}_V^N$ . Finally, we claim that  $s'$  is a closed immersion:

Notice in the above diagram, the tall rectangle involving  $\mathbb{P}_V^m \times_V \mathbb{P}_V^n, \mathbb{P}_{\mathbb{Z}}^m \times_{\text{Spec}(\mathbb{Z})} \mathbb{P}_{\mathbb{Z}}^n, V, \text{Spec}(\mathbb{Z})$  forms a fibre square by suitable diagram chase.

Then, this in fact enforces  $\mathbb{P}_V^m \times_V \mathbb{P}_V^n \cong \mathbb{P}_V^N \times_{\mathbb{P}_{\mathbb{Z}}^N} (\mathbb{P}_{\mathbb{Z}}^m \times_{\text{Spec}(\mathbb{Z})} \mathbb{P}_{\mathbb{Z}}^n)$ : Given any scheme  $W$  together with two morphisms  $w_1 : W \rightarrow \mathbb{P}_V^N$  and  $w_2 : W \rightarrow \mathbb{P}_{\mathbb{Z}}^m \times_{\text{Spec}(\mathbb{Z})} \mathbb{P}_{\mathbb{Z}}^n$ , such that the following holds:

$$\begin{array}{ccc} W & \xrightarrow{w_2} & \mathbb{P}_{\mathbb{Z}}^m \times_{\text{Spec}(\mathbb{Z})} \mathbb{P}_{\mathbb{Z}}^n \\ \downarrow \exists! w & & \downarrow s \\ \mathbb{P}_V^m \times_V \mathbb{P}_V^n & \xrightarrow{p} & \mathbb{P}_{\mathbb{Z}}^m \times_{\text{Spec}(\mathbb{Z})} \mathbb{P}_{\mathbb{Z}}^n \\ \downarrow \exists! s' & & \downarrow s \\ \mathbb{P}_V^N & \xrightarrow{\quad} & \mathbb{P}_{\mathbb{Z}}^N \\ \downarrow \pi_{V,N} & & \downarrow \\ V & \xrightarrow{\quad} & \text{Spec}(\mathbb{Z}) \end{array}$$

$w_1$

Which, the existence and uniqueness of  $w : W \rightarrow \mathbb{P}_V^m \times_V \mathbb{P}_V^n$  is guaranteed by the tall fibre rectangle, showing the top square is a fibre square. Since this characterizes  $\mathbb{P}_V^m \times_V \mathbb{P}_V^n$  as the fibre product  $\mathbb{P}_V^N \times_{\mathbb{P}_{\mathbb{Z}}^N} (\mathbb{P}_{\mathbb{Z}}^m \times_{\text{Spec}(\mathbb{Z})} \mathbb{P}_{\mathbb{Z}}^n)$ , then  $s' : \mathbb{P}_V^m \times_V \mathbb{P}_V^n \rightarrow \mathbb{P}_V^N$  is a base change of  $s : \mathbb{P}_{\mathbb{Z}}^m \times_{\text{Spec}(\mathbb{Z})} \mathbb{P}_{\mathbb{Z}}^n \hookrightarrow \mathbb{P}_{\mathbb{Z}}^N$  (by the morphism  $\mathbb{P}_V^N \rightarrow \mathbb{P}_{\mathbb{Z}}^N$ ), showing  $s'$  is a closed immersion (since  $s$  is).

Finally, we get this massive commutative diagram:

$$\begin{array}{ccccccc}
T & \xrightarrow{\iota_T} & \mathbb{P}_U^n & \xrightarrow{(\iota_U)'} & \mathbb{P}_{\mathbb{P}_V^m}^n \cong \mathbb{P}_V^m \times_V \mathbb{P}_V^n & \xrightarrow{s'} & \mathbb{P}_V^N \\
\rho \downarrow & \swarrow \pi_{U,n} & & \swarrow (\pi_{V,n})' & & & \\
U & \xrightarrow{\iota_U} & \mathbb{P}_V^m & & & & \\
\varphi \downarrow & \swarrow \pi_{V,m} & & \searrow \pi_{V,N} & & & \\
V & & & & & & 
\end{array}$$

Which,  $\iota_T, (\iota_U)', s'$  are all constructed as closed immersions, hence the composition  $T \hookrightarrow \mathbb{P}_V^N$  is also a closed immersion. This realizes  $\varphi \circ \rho : T \rightarrow V$  as a projective morphism, hence composition of projective morphisms are still projective.

Finally, back to the original problem, given  $f : X \rightarrow S$  and  $g : Y \rightarrow S$  two projective morphisms, (a) guarantees their pullbacks to be projective morphisms also. So,  $f' : X \times_S Y \rightarrow Y$  and  $g' : X \times_S Y \rightarrow X$  are projective morphisms, satisfying the following fibre square:

$$\begin{array}{ccc}
X \times_S Y & \xrightarrow{f'} & Y \\
g' \downarrow & & \downarrow g \\
X & \xrightarrow{f} & S
\end{array}$$

Then, (b) guarantees the composition of projective morphisms to be projective, hence  $f \circ g' = g \circ f' : X \times_S Y \rightarrow S$  is projective, finishing our statement.



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### 3

**Problem 3.** Let  $f : X \rightarrow Y$  be a morphism of separated scheme of finite type over a Noetherian scheme  $S$ . Let  $Z \subset X$  be a closed subscheme which is proper over  $S$ . Show that  $f(Z)$  is closed in  $Y$ .

---

*Solution.* To show  $f(Z) \subseteq Y$  is closed, we'll show that with the closed immersion  $\iota : Z \hookrightarrow X$ , one has  $f \circ \iota : Z \rightarrow Y$  is proper, which will show  $f \circ \iota(Z) = f(Z)$  is closed in  $Y$ .

First, let  $\varphi_X : X \rightarrow S$  and  $\varphi_Y : Y \rightarrow S$  be the morphism that define  $X, Y$  as separated schemes of finite type over  $S$ , this implies both  $\varphi_X, \varphi_Y$  are of finite type and separated morphisms.

Also, consider the closed immersion  $\iota : Z \hookrightarrow X$ , since  $Z$  is proper over  $S$ , we have this morphism  $\varphi_X \circ \iota : Z \rightarrow S$  be proper.

Now, since  $f : X \rightarrow Y$  is a morphism of schemes over  $S$ , we have the following commutative diagram:

$$\begin{array}{ccccc} & & Z & & \\ & & \searrow & & \\ & & \iota & & \\ & & \searrow & & \\ & X & \xrightarrow{f} & Y & \\ & \searrow & & \swarrow & \\ & \varphi_X & & \varphi_Y & \\ & & S & & \end{array}$$

Or,  $\varphi_X = \varphi_Y \circ f$ , which implies  $\varphi_X \circ \iota = \varphi_Y \circ (f \circ \iota)$ . Since  $\varphi_X \circ \iota$  is proper, with  $\varphi_Y$  being separated, we have  $f \circ \iota$  being proper also (which is a closed map). With  $Z$  being closed in itself,  $f \circ \iota(Z) = f(Z) \subseteq Y$  is closed. This finishes our proof.

## 4

**Problem 4.** Let  $k$  be a field and let  $R = k[X, Y, Z]/I$ , where  $I = (X - YZ, XZ - Y^2)$ . Let  $W = \text{Spec}(R)$ .

- Is  $W$  a reduced scheme? Justify your answer.
- Is  $W$  irreducible? If not, what are its irreducible components?
- Prove or disprove that  $W$  has infinitely many rational points if  $k$  is infinite (recall that a point  $x \in W$  is called rational point if the canonical map  $k \rightarrow k(x)$  is an isomorphism, where  $k(x)$  is the residue field of the local ring  $\mathcal{O}_{W,x}$ ).

*Solution.* For the sake of working through all subproblems, we'll first do some calculations regarding two ideals of  $k[X, Y, Z]$ , defined by  $P = (X, Y)$  and  $Q = (X - Z^3, Y - Z^2)$ . In particular, we claim that they're prime ideals that contains  $I$ :

- For  $P = (X, Y)$ , it's clear that  $k[X, Y, Z]/P \cong k[Z]$ , hence it's prime; on the other hand, one has  $X, Y \in P$  implies  $YZ, XZ, Y^2 \in P$ , hence  $X - YZ, XZ - Y^2 \in P$ , showing  $I \subseteq P$ .
- For  $Q$ , we'll show that  $k[X, Y, Z]/Q \cong k[T]$  also: Consider the ring homomorphism  $\varphi : k[X, Y, Z] \rightarrow k[T]$  by  $\varphi(f(X, Y, Z)) = f(T^3, T^2, T)$ . Which, this morphism is surjective, as any polynomial  $f(T) \in k[T]$  also appears in  $k[X, Y, Z]$  (say denotes as  $f(Z)$ ), then  $\varphi(f(Z)) = f(T)$ . This shows that  $k[X, Y, Z]/\ker(\varphi) \cong k[T]$ .

Now, we show that  $Q = \ker(\varphi)$ : For the generators of  $Q$ , we have  $\varphi(X - Z^3) = T^3 - (T)^3 = 0$ , and  $\varphi(Y - Z^2) = T^2 - (T)^2 = 0$ , hence  $Q \subseteq \ker(\varphi)$  (as its generators are contained in their kernel).

For the converse, we claim that all polynomials can be expressed in the form  $f \cdot (X - Z^3) + g \cdot (Y - Z^2) + h(Z)$  for some  $f, g \in k[X, Y, Z]$ , and  $h(Z) \in k[Z]$ : Given any monomial  $X^n Y^m Z^p$ , we can rewrite it as the following:

$$\begin{aligned} X^n Y^m Z^p &= ((X - Z^3) + Z^3)^n ((Y - Z^2) + Z^2)^m Z^p \\ &= \left( \sum_{k=0}^n \binom{n}{k} (X - Z^3)^k (Z^3)^{n-k} \right) \left( \sum_{l=0}^m \binom{m}{l} (Y - Z^2)^l (Z^2)^{m-l} \right) Z^p \\ &= \left( Z^{3n} + \sum_{k=1}^n \binom{n}{k} (X - Z^3)^k (Z^3)^{n-k} \right) \left( Z^{2m} + \sum_{l=1}^m (Y - Z^2)^l (Z^2)^{m-l} \right) Z^p \end{aligned} \quad (4.1)$$

Which, define  $f := \sum_{k=1}^n \binom{n}{k} (X - Z^3)^{k-1} (Z^3)^{n-k}$ , and  $g := \sum_{l=1}^m (Y - Z^2)^{l-1} (Z^2)^{m-l}$ , the above can be rewritten as follow:

$$\begin{aligned} X^n Y^m Z^p &= (Z^{3n} + f \cdot (X - Z^3))(Z^{2m} + g \cdot (Y - Z^2))Z^p \\ &= (f(Z^{2m} + g(Y - Z^2))Z^p) \cdot (X - Z^3) + (gZ^{3n+p}) \cdot (Y - Z^2) + Z^{3n+2m+p} \end{aligned} \quad (4.2)$$

This shows that each monomial can be written as some  $f \cdot (X - Z^3) + g \cdot (Y - Z^2) + Z^q$ . Hence, with all polynomials being finite  $k$ -linear combinations of the monomials, they can all be written in the form of  $f \cdot (X - Z^3) + g \cdot (Y - Z^2) + h(Z)$  for  $f, g \in k[X, Y, Z]$  and  $h(Z) \in k[Z]$ .

As a consequence, for any polynomial  $\ell(X, Y, Z) \in \ker(\varphi)$ , since there exists  $f, g \in k[X, Y, Z]$  and  $h(Z) \in k[Z]$ , such that  $\ell(X, Y, Z) = f \cdot (X - Z^3) + g \cdot (Y - Z^2) + h(Z)$ , then the evaluation by  $\varphi$  gives the following:

$$\varphi(\ell(X, Y, Z)) = \varphi(f \cdot (X - Z^3)) + \varphi(g \cdot (Y - Z^2)) + \varphi(h(Z)) = h(T) = 0 \quad (4.3)$$

(Note: recall that  $X - Z^3, Y - Z^2 \in \ker(\varphi)$  are verified). Hence, one must have  $h(T) = 0$ , or  $h(Z) = 0$ , showing that  $\ell(X, Y, Z) = f \cdot (X - Z^3) + g \cdot (Y - Z^2) \in Q$ . This proves that  $\ker(\varphi) \subseteq Q$ , finishing the proof of  $\ker(\varphi) = Q$ .

Then, since  $k[X, Y, Z]/Q \cong k[T]$  based on the homomorphism  $\varphi$ , then  $Q$  is a prime ideal.

Finally, to say  $I \subseteq Q$ , use the algorithm proposed above, some calculation shows the following:

$$\begin{aligned}
 \bullet \quad X - YZ &= X - ((Y - Z^2) + Z^2)Z = (X - Z^3) + Z(Y - Z^2) \in Q & (4.4) \\
 \bullet \quad XZ - Y^2 &= XZ - ((Y - Z^2) + Z^2)^2 \\
 &= XZ - ((Y - Z^2)^2 + 2Z^2(Y - Z^2) + Z^4) \\
 &= XZ - Z^4 - ((Y - Z^2) + 2Z^2)(Y - Z^2) \\
 &= Z(X - Z^3) + (Y + Z^2)(Y - Z^2) \in Q
 \end{aligned} \tag{4.5}$$

Hence, the generators of  $I$  are in  $Q$ , showing  $I \subseteq Q$ .

Now, since  $W = \text{Spec}(R) \cong V(I)$  as closed subschemes of  $\text{Spec}(k[X, Y, Z])$ , we'll work in this regime.  
**Here are the solutions to the subproblems:**

- (a) Here, we claim that  $W$  is reduced, by proving that  $R$  is reduced. In particular, we'll show that  $I = P \cap Q$  to show that it's a radical.

It's clear that  $I \subseteq P \cap Q$  based on the previous calculation. To show that  $P \cap Q \subseteq I$ , we'll consider the algorithm used when proving  $k[X, Y, Z]/Q \cong k[T]$ .

First, for any polynomial  $s \in P \cap Q$ , since  $s \in P = (X, Y)$ , so it can be written in the form  $s = v_1X + v_2Y$  for some  $v_1, v_2 \in k[X, Y, Z]$ .

Then, perform the algorithm for  $Q$  on  $v_1, v_2$ , there exists  $f_1, f_2, g_1, g_2 \in k[X, Y, Z]$ , and  $h_1(Z), h_2(Z) \in k[Z]$ , such that the following holds:

$$\begin{aligned}
 s &= v_1X + v_2Y \\
 &= (f_1(X - Z^3) + g_1(Y - Z^2) + h_1(Z))X + (f_2(X - Z^3) + g_2(Y - Z^2) + h_2(Z))Y \tag{4.6} \\
 &= f_1(X - Z^3)X + g_1(Y - Z^2)X + f_2(X - Z^3)Y + g_2(Y - Z^2)Y + h_1(Z)X + h_2(Z)Y
 \end{aligned}$$

Which, the first four terms in the sum is in  $PQ$  (since they're multiples of products of the generators of  $P = (X, Y), Q = (X - Z^3, Y - Z^2)$ ). So, in case for  $s \in P \cap Q$ , one needs  $h_1(Z)X + h_2(Z)Y \in P \cap Q$ .

It's true that  $h_1(Z)X + h_2(Z)Y \in P$ , hence the only condition is  $h_1(Z)X + h_2(Z)Y \in Q$ . Which, we can rewrite it as follow:

$$\begin{aligned}
 h_1(Z)X + h_2(Z)Y &= h_1(Z)((X - Z^3) + Z^3) + h_2(Z)((Y - Z^2) + Z^2) \\
 &= h_1(Z)(X - Z^3) + h_2(Z)(Y - Z^2) + (h_1(Z)Z^3 + h_2(Z)Z^2)
 \end{aligned} \tag{4.7}$$

Which, one requires  $h_1(Z)Z^3 + h_2(Z)Z^2 \in Q = (X - Z^3, Y - Z^2)$ . Then, this imposes that  $h_1(Z)Z^3 + h_2(Z)Z^2 = 0$  (as it's independent from  $X, Y$ ). So, one has  $Z^2(h_1(Z)Z + h_2(Z)) = 0$ , showing  $h_2(Z) = -h_1(Z)Z$ .

As a result, the term  $h_1(Z)X + h_2(Z)Y$  becomes the following:

$$h_1(Z)X - h_1(Z)YZ = h_1(Z)(X - YZ) \in I \tag{4.8}$$

So, this shows that the last two terms in  $s = f_1(X - Z^3)X + g_1(Y - Z^2)X + f_2(X - Z^3)Y + g_2(Y - Z^2)Y + h_1(Z)X + h_2(Z)Y$  is contained in  $I$ . To show  $s \in I$ , one simply needs the first four terms to be in  $I$  also.

Since  $PQ = (X, Y)(X - Z^3, Y - Z^2) = ((X - Z^3)X, (Y - Z^2)X, (X - Z^3)Y, (Y - Z^2)Y)$ , it suffices to prove these generators are in  $I$ :

$$\begin{aligned} \bullet \quad (X - Z^3)X &= X^2 - (XZ)Z^2 = X^2 - (XZ)Z^2 + -Y^2Z^2 + Y^2Z^2 \\ &= (X^2 - (YZ)^2) - Z^2(XZ - Y^2) \\ &= (X + YZ)(X - YZ) = Z^2(XZ - Y^2) \in I \end{aligned} \quad (4.9)$$

$$\begin{aligned} \bullet \quad (Y - Z^2)X &= XY - (XZ)Z = XY - (XZ)Z + Y^2Z - Y^2Z \\ &= Y(X - YZ) - Z(XZ - Y^2) \in I \end{aligned} \quad (4.10)$$

$$\begin{aligned} \bullet \quad (X - Z^3)Y &= XY - YZ^3 + XZ^2 - XZ^2 \\ &= Z^2(X - YZ) + XY - XZ^2 - Y^2Z + Y^2Z \\ &= Z^2(X - YZ) + Y(X - YZ) - Z(XZ - Y^2) \\ &= (Z^2 + Y)(X - YZ) - Z(XZ - Y^2) \in I \end{aligned} \quad (4.11)$$

$$\begin{aligned} \bullet \quad (Y - Z^2)Y &= Y^2 - YZ^2 - XZ + XZ \\ &= Z(X - YZ) - (XZ - Y^2) \in I \end{aligned} \quad (4.12)$$

This shows the generators of  $PQ$  are in  $I$ , hence  $PQ \subseteq I$ . As a consequence, the first four terms of  $s = f_1(X - Z^3)X + g_1(Y - Z^2)X + f_2(X - Z^3)Y + g_2(Y - Z^2)Y + h_1(Z)X + h_2(Z)Y$  is also in  $I$ .

This proves  $s \in I$ , showing  $P \cap Q \subseteq I$ . Hence,  $I = P \cap Q$ , showing it's a radical. This shows that  $R = k[X, Y, Z]/I$  is reduced, hence  $W = \text{Spec}(R)$  is also reduced.

- (b) We show that  $W$  is reducible, in particular that  $W = V(I) = V(P) \cup V(Q)$ , where  $V(P), V(Q)$  are its irreducible components.

First, since  $I \subseteq P$  and  $I \subseteq Q$ , it's clear that  $V(P), V(Q) \subseteq V(I)$ , hence  $V(P) \cup V(Q) \subseteq V(I)$ ; to claim the reverse inclusion, we'll consider the following element:

$$Z(X - YZ) - (XZ - Y^2) = XZ - YZ^2 - XZ + Y^2 = Y(Y - Z^2) \in I \quad (4.13)$$

Suppose  $K \in V(I)$  is a prime ideal containing  $I$ , then  $Y(Y - Z^2) \in K$ , which the prime property guarantees  $Y \in K$  or  $(Y - Z^2) \in K$ .

- If  $Y \in K$ , then since  $X - YZ \in I \subseteq K$ , one has  $X = (X - YZ) + YZ \in K$ , showing  $P = (X, Y) \subseteq K$ , or  $K \in V(P)$ .
- Else if  $Y - Z^2 \in K$ , then since  $X - YZ \in I \subseteq K$  again, one has  $X - Z^3 = (X - YZ) + (YZ - Z^3) = (X - YZ) + Z(Y - Z^2) \in K$ , showing  $Q = (X - Z^3, Y - Z^2) \subseteq K$ , or  $K \in V(Q)$ .

Hence, in either case  $K \in V(P) \cup V(Q)$ , showing  $W = V(I) = V(P) \cup V(Q)$ .

To show that  $W$  is reducible, we'll show that  $V(P), V(Q)$  are proper closed subsets of  $W$ : Suppose the contrary that one of them is not proper closed subset of  $W$ , there are two cases:

- If  $V(P) = W$ , this shows that  $V(Q) \subseteq V(P)$ ; with  $P, Q$  being prime (which both are radicals), this shows that  $Q \supseteq P$ . However, if consider the ring homomorphism  $\psi: k[X, Y, Z] \twoheadrightarrow k$  by  $\psi(X, Y, Z) = \psi(1, 1, 1)$ , notice that  $\psi(X - Z^3) = 1 - 1^3 = 0$  and  $\psi(Y - Z^2) = 1 - 1^2 = 0$ , showing  $Q = (X - Z^3, Y - Z^2) \subseteq \ker(\psi)$ ; yet, if plugin the generators of  $P$ , we get  $\psi(X) = \psi(Y) = 1 \neq 0$ , showing  $P \not\subseteq \ker(\psi)$ , which contradicts our construction that  $\ker(\varphi) \supseteq Q \supseteq P$ .

- Else if  $V(Q) = W$ , this shows that  $V(P) \subseteq V(Q)$ , which implies  $P \supseteq Q$ . However, consider the ring homomorphism  $\rho : k[X, Y, Z] \twoheadrightarrow k$  by  $\rho(f(X, Y, Z)) = f(0, 0, 1)$ , notice that  $\rho(X) = \rho(Y) = 0$ , showing  $P = (X, Y) \subseteq \ker(\rho)$ ; yet, if plugin the generators of  $Q$ , we get  $\rho(X - Z^3) = 0 - 1^3 = -1 \neq 0$ , showing  $Q \not\subseteq \ker(\rho)$ , again this contradicts our construction that  $\ker(\rho) \supseteq P \supseteq Q$ .

Since in either case we get a contradiction, then  $V(P), V(Q)$  cannot be the whole  $W$ , showing they're proper closed subsets, which proves that  $W = V(P) \cup V(Q)$  is reducible.

Finally, to show  $V(P), V(Q)$  are its irreducible components, consider any other irreducible subsets  $W' \subseteq W = V(I)$ , since  $W'$  is also closed in  $\text{Spec}(k[X, Y, Z])$  (by the closeness of  $V(I)$  in it), then  $W' = V(K)$  for some ideal  $K \subseteq k[X, Y, Z]$ . Which, the irreducibility of  $W'$  guarantees  $K$  to be a prime ideal. However, if  $V(K) \subseteq V(I)$ , one has  $I \subseteq \sqrt{I} \subseteq K$  (by primeness, it is a radical), showing that  $K \in V(I) = V(P) \cup V(Q)$ . This implies that either  $K \in V(P)$  (implying  $P \subseteq K$ , or  $V(P) \supseteq V(K)$ ), or  $K \in V(Q)$  (implying  $Q \subseteq K$ , or  $V(Q) \supseteq V(K)$ ), showing that  $W' = V(K)$  is contained in  $V(P)$  or  $V(Q)$ . Hence, these two must be the irreducible components of  $W$ .

- (c) With  $k$  being infinite, we'll show that there are infinitely rational points in  $W$ . For any  $t \in k$ , consider the maximal ideal  $\mathfrak{m}_t := (X, Y, Z - t) \subset k[X, Y, Z]$ . Notice that it contains  $I$ , since  $X - YZ, XZ - Y^2 \in \mathfrak{m}_t$  by the fact that  $X, Y \in \mathfrak{m}_t$ . Hence,  $x := \mathfrak{m}_t$  can be realized as a maximal ideal in  $R = k[X, Y, Z]/I$  by descending to the quotient, which is realized as a closed point in  $W = \text{Spec}(R)$ .

To compute its residue field, let  $\overline{\mathfrak{m}}_t \subseteq R$  denotes the quotient of  $\mathfrak{m}_t$ . Then, its corresponding residue field is as follow:

$$k(\overline{\mathfrak{m}}_t) = \frac{R_{\overline{\mathfrak{m}}_t}}{\overline{\mathfrak{m}}_t R_{\overline{\mathfrak{m}}_t}} \cong \left( \frac{R}{\overline{\mathfrak{m}}_t} \right)_{\overline{\mathfrak{m}}_t} \cong k \quad (4.14)$$

Note that since  $I \subseteq \mathfrak{m}_t$ , then the composition of projections  $\varphi : k[X, Y, Z] \twoheadrightarrow R \twoheadrightarrow R/\overline{\mathfrak{m}}_t$  has the kernel being precisely  $\varphi^{-1}(\overline{\mathfrak{m}}_t) = \mathfrak{m}_t$ , hence  $k \cong k[X, Y, Z]/\mathfrak{m}_t \cong R/\overline{\mathfrak{m}}_t$  (where, the first isomorphism can be realized by the evaluation  $e : k[X, Y, Z] \twoheadrightarrow k$  by  $e(f(X, Y, Z)) = f(0, 0, t)$ , where the kernel is precisely  $\mathfrak{m}_t = (X, Y, Z - t)$ ).

Hence, with  $k$  being infinite, there are infinitely many  $t \in k$ , with  $\overline{\mathfrak{m}}_t \subseteq R$  being a maximal ideal (or a closed point in  $W = \text{Spec}(R)$ ), such that the residue field is isomorphic to  $k$ . Hence,  $W$  has infinitely many rational points.

## 5

**Problem 5.** Let  $f : X \rightarrow Y$  be a surjective morphism between two schemes which are finite type and separated integral schemes over a field. Assume that  $X$  is an affine scheme. Prove or disprove that  $Y$  is an affine scheme if and only if it is a quasi-affine scheme (recall that quasi-affine means open subscheme of an affine scheme).

*Solution.* We'll prove that  $Y$  is affine  $\iff Y$  is quasi-affine.

$\implies$ :

Suppose  $Y$  is affine, then it's an open subscheme of  $Y$  (an affine scheme), hence  $Y$  is quasi-affine.

$\impliedby$ :

Suppose that  $Y$  is quasi-affine. Here, we'll denote the global section of  $Y$  as  $A := \mathcal{O}_Y(Y)$ , and let  $B$  be a commutative ring satisfies  $X \cong \text{Spec}(B)$  (which,  $X, Y$  being integral schemes implies  $A, B$  are integral domains).

Let's first deal with a special case: Suppose  $B$  is a field, then  $X = \text{Spec}(B)$  is a one point scheme, hence with  $f : X \rightarrow Y$  being surjective,  $Y$  must be a one point scheme also. Take any affine neighborhood of the only point of  $Y$ , it must be  $Y$  itself, showing that  $Y = \text{Spec}(A')$  for some ring  $A'$  (in particular,  $A' = \mathcal{O}_Y(Y) = A$ ). So,  $Y$  is affine.

From now on, we'll assume  $B$  (the global section of  $X$ ) is not a field.

Let  $k$  be a field where  $X, Y$  are finite type and separated integral schemes over it. Which, given  $\varphi_X : X \rightarrow \text{Spec}(k)$ ,  $\varphi_Y : Y \rightarrow \text{Spec}(k)$  as the corresponding morphisms,  $\varphi_X, \varphi_Y$  are finite type and separated morphisms that satisfy the following diagram:

$$\begin{array}{ccc} X = \text{Spec}(B) & \xrightarrow{f} & Y \\ & \searrow \varphi_X & \swarrow \varphi_Y \\ & \text{Spec}(k) & \end{array}$$

Then, on the global section level, this reverses to the following diagram of ring homomorphisms:

$$\begin{array}{ccc} B & \xleftarrow{f^\#} & \mathcal{O}_Y(Y) = A \\ & \swarrow \varphi_X^\# & \searrow \varphi_Y^\# \\ & k & \end{array}$$

Now, we claim the following statements in order:

- (a)  $Y$  can be realized as open subschemes of  $\text{Spec}(A)$ .
- (b)  $Y$  is affine, in particular  $Y \cong \text{Spec}(A)$ .

### Proof of (a):

Since  $Y$  is assumed to be quasi-affine, there exists an open immersion  $\iota_Y : Y \hookrightarrow \text{Spec}(R)$  for some commutative ring  $R$ . Then, since the affinization of  $Y$ , say  $\text{Aff}_Y = \text{Spec}(A)$ , with the natural morphism of schemes  $i_Y : Y \rightarrow \text{Aff}_Y$ , there exists a unique morphism of schemes  $\psi_Y : \text{Aff}_Y \rightarrow \text{Spec}(R)$ , such that the following commutative diagram holds:

$$\begin{array}{ccc}
Y & \xrightarrow{i_Y} & \text{Aff}_Y \\
\searrow \iota_Y & & \swarrow \exists! \psi_Y \\
& \text{Spec}(R) &
\end{array}$$

Which, since  $\iota_Y$  is injective (as set map), this enforces  $i_Y$  to also be injective also.

Now, consider the ring homomorphism  $\psi_Y^\# : R \rightarrow A$ . For any  $P \in Y \hookrightarrow \text{Spec}(R)$ , there exists some element  $f \in R$ , such that  $P \in D(f) \subseteq Y$  (since  $Y$  is open subscheme of  $\text{Spec}(R)$ ). Then, denote  $f' := \varphi_Y^\#(f) \in A$ . Using the Hartshorne Notation, we can consider the subset  $Y_{f'} \subseteq Y$  (denoting all points  $y \in Y$ , such that the localization  $f'_y \in \mathcal{O}_{Y,y}$  is invertible, or not in the maximal ideal  $\mathfrak{m}_y \subseteq \mathcal{O}_{Y,y}$ ). Notice that this set is precisely  $Y_{f'} = D(f)$ , since when passing to the stalk, any  $P \in D(f)$  has  $\mathcal{O}_{Y,P}$  has  $f$  being an invertible element, hence with  $f'$  being the image of  $f$  in  $A$ , under localization it's also invertible.

As a result, one has  $R_f \cong \mathcal{O}_{\text{Spec}(R)}(D(f)) = \mathcal{O}_Y(D(f)) = \mathcal{O}_Y(Y_{f'}) = \mathcal{O}_Y(D(f')) \cong A_{f'}$ , showing that  $P \in D(f) \subseteq Y$  has an affine neighborhood  $D(f) \cong D(f') \cong \text{Spec}(A_{f'})$  in  $Y$ .

This shows that  $i_Y : Y \hookrightarrow \text{Spec}(A)$  is actually an open immersion (as any point in  $Y$  has an open neighborhood in  $Y$ , such that it's scheme-isomorphic to a fundamental open subset of  $\text{Spec}(A)$ ). So,  $Y$  is an open subscheme of  $\text{Spec}(A)$ .

### Proof of (b):

Now, together with the open immersion  $i_Y : Y \hookrightarrow \text{Spec}(A)$ , the morphism  $\varphi_Y : Y \rightarrow \text{Spec}(k)$  uniquely factors through some morphism  $\rho : \text{Spec}(A) \rightarrow \text{Spec}(k)$  (since  $\text{Spec}(A) = \text{Aff}_Y$  the affinization), which we have the following commutative diagram of morphisms:

$$\begin{array}{ccccc}
X & \xrightarrow{f} & Y & \xrightarrow{i_Y} & \text{Spec}(A) \\
\searrow \varphi_X & & \downarrow \varphi_Y & & \swarrow \exists! \rho \\
& & \text{Spec}(k) & &
\end{array}$$

Where the morphism  $i_Y \circ f$  is separated (as it's morphism between affine schemes). Which, on the level of global section, we have the following commutative diagram:

$$\begin{array}{ccccc}
B & \xleftarrow{f^\#} & A & \xleftarrow{\text{id}_A} & A \\
\swarrow \varphi_X^\# & & \uparrow \varphi_Y^\# & & \searrow \rho^\# \\
& & k & &
\end{array}$$

Where the middle  $A$  is given by the global section of  $Y$ ,  $A = \mathcal{O}_Y(Y)$ .

Then, this implies that the morphism  $i_Y \circ f : X = \text{Spec}(B) \rightarrow \text{Spec}(A)$  is induced by the ring homomorphism  $f^\# \circ \text{id}_A = f^\# : A \rightarrow B$ ; and, for the diagram to commute,  $f^\#$  is in fact a  $k$ -algebra homomorphism. Moreover,  $f^\#$  is of finite type, as  $B$  is a finitely generated  $k$ -algebra, and the morphism is a  $k$ -algebra homomorphism. This realizes  $B$  as a finitely generated  $A$ -algebra also.

Now, notice that  $i_Y \circ f$  is a dominant morphism: Because  $A$  is an integral domain, then  $\text{Spec}(A)$  is irreducible, hence  $Y \subseteq \text{Spec}(A)$  as an open subscheme is dense and irreducible (or  $\overline{Y} = \text{Spec}(A)$ ). With  $f$  being surjective, one has  $f(X) = Y$ , hence  $\overline{i_Y \circ f(X)} = \overline{i_Y(Y)} = \overline{Y} = \text{Spec}(A)$ . Hence, with the morphism  $i_Y \circ f$  being dominant, the induced ring homomorphism  $f^\#$  is injective.

Finally, we claim that  $Y = \text{Spec}(A)$ : Since  $Y \subseteq \text{Spec}(A)$  is open, there exists  $a_i \in A$  (where  $i \in I$  for some index set), such that  $Y = \bigcup_{i \in I} D(a_i)$ . Since  $\varphi_Y : Y \rightarrow \text{Spec}(k)$  is of finite type (which is quasi-compact), with  $\text{Spec}(k)$  being quasi-compact, this implies  $Y = \varphi_Y^{-1}(\text{Spec}(k))$  is quasi-compact. Hence, one can choose the cover to be finite, say  $a_1, \dots, a_n \in A$  satisfies  $Y = \bigcup_{i=1}^n D(a_i)$ .

Then, with  $f : X \twoheadrightarrow Y$  and the induced ring homomorphism  $f^\# : A \hookrightarrow B$ , one has  $\text{Spec}(B) = X = f^{-1}(Y) = \bigcup_{i=1}^n f^{-1}(D(a_i)) = \bigcup_{i=1}^n D(f^\#(a_i))$ . So, since  $D(f^\#(a_i))$  forms an open cover of  $\text{Spec}(B)$ , this implies  $B = (f^\#(a_1), \dots, f^\#(a_n))$ .

At this point I was aiming for  $A = (a_1, \dots, a_n)$  (currently in progress), then with  $\bigcup_{i=1}^n D(a_i) = \text{Spec}(A)$  based on this argument, it'll finish the proof that  $Y = \text{Spec}(A)$ , or showing  $Y$  is affine.